

1. Matrix Notation.

- (a) Show that $A^*A = I$ if and only if the columns of A are orthonormal.
- (b) Show that a vector $\mathbf{x}$ is orthogonal to every column of A if and only if $A^*\mathbf{x} = \mathbf{0}$.
- (c) Suppose that $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ for some vectors $\mathbf{v}_i$ and scalars λ_i . Show how to express this with a single matrix equation $AV = V\Lambda$.
- (d) Suppose that $A = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ for some vectors $\mathbf{u}_i, \mathbf{v}_i$ and scalars σ_i . Show how to express this with a single matrix equation $A = U\Sigma V^*$.

2. Projection. Let $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ with $\langle \mathbf{v}, \mathbf{u} \rangle = \mathbf{v}^*\mathbf{u} \neq 0$ and consider the $n \times n$ matrix

$$P = \frac{1}{\mathbf{v}^*\mathbf{u}} \mathbf{u}\mathbf{v}^*.$$

- (a) Show that $P^2 = P$. [Hint: Use associativity of matrix multiplication.]
- (b) Show that $P(t\mathbf{u}) = t\mathbf{u}$ for all scalars $t \in \mathbb{C}$.
- (c) If $\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^*\mathbf{w} = 0$, show that $P\mathbf{w} = \mathbf{0}$.
- (d) If $\mathbf{v} = \mathbf{u}$, show that the linear function $\mathbb{C}^n \rightarrow \mathbb{C}^n$ defined by $\mathbf{x} \mapsto P\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto the subspace (complex line) $\{t\mathbf{u} : t \in \mathbb{C}\}$.

3. Special Matrices. For any real number t , consider the following three real 2×2 matrices:

$$R_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \quad F_t = \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}, \quad P_t = \begin{pmatrix} \cos^2 t & \cos t \sin t \\ \cos t \sin t & \sin^2 t \end{pmatrix}.$$

- (a) Describe what each matrix does geometrically. [Hint: Rotate, reFlect, Project.]
- (b) Use trig identities to prove the following matrix identities:

$$F_t^2 = I, \quad P_t^2 = P_t, \quad F_{2t} + I = 2P_t, \quad R_s R_t = R_{s+t}, \quad F_s F_t = R_{s-t}.$$

- (c) Give geometric explanations for these identities.

4. Unitary and Orthogonal Matrices. A complex square matrix $U \in \mathbb{C}^{n \times n}$ is called *unitary* if $U^*U = I$. Unitary matrices with real entries are called *orthogonal* matrices. [In this case we have $U^T U = I$ because $U^* = U^T$ for real matrices.]

- (a) Show that a product of unitary matrices is unitary.
- (b) Show that unitary matrices preserve distance. That is, prove that we have $\|U\mathbf{x} - U\mathbf{y}\| = \|\mathbf{x} - \mathbf{y}\|$ for all unitary matrices $U \in \mathbb{C}^{n \times n}$ and vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$.
- (c) Show that the matrices R_t and F_t from the previous problem are orthogonal.
- (d) Show that any 2×2 orthogonal matrix is equal to R_t or F_t for some t . [Hint: Let $U \in \mathbb{R}^{2 \times 2}$ have columns $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^2$. If $U^T U = I$ then $\mathbf{u}_1, \mathbf{u}_2$ is an orthonormal set. If $\|\mathbf{u}_1\| = 1$ then we can write $\mathbf{u}_1 = (\cos t, \sin t)$ for some unique angle $0 \leq t < 2\pi$.]

5. Frobenius Norm. For any complex matrix $A = (a_{ij})$ we define the Frobenius norm by

$$\|A\|_F := \sqrt{\sum_{i,j} |a_{ij}|^2}.$$

- (a) Let $\|\cdot\|$ denote the standard Hermitian norm on complex vectors. If $\mathbf{a}_1, \dots, \mathbf{a}_m$ are the columns and $\mathbf{r}_1^*, \dots, \mathbf{r}_\ell^*$ are the rows of a matrix $A \in \mathbb{C}^{\ell \times m}$, show that

$$\|A\|_F^2 = \|\mathbf{a}_1\|^2 + \dots + \|\mathbf{a}_m\|^2 = \|\mathbf{r}_1\|^2 + \dots + \|\mathbf{r}_\ell\|^2.$$

- (b) For all matrices $A \in \mathbb{C}^{\ell \times m}$ and vectors $\mathbf{x} \in \mathbb{C}^m$, show that

$$\|A\mathbf{x}\| \leq \|A\|_F \|\mathbf{x}\|.$$

[Hint: If $\mathbf{r}_i^*$ is the i th row of A , then $\mathbf{r}_i^* \mathbf{x}$ is the i th entry of $A\mathbf{x}$. From Cauchy-Schwarz we know that $|\mathbf{r}_i^* \mathbf{x}|^2 \leq \|\mathbf{r}_i^*\|^2 \|\mathbf{x}\|^2$. Now use the row formula from part (a).]

- (c) For any matrices $A \in \mathbb{C}^{\ell \times m}$ and $B \in \mathbb{C}^{m \times n}$, combine parts (a) and (b) to prove that

$$\|AB\|_F \leq \|A\|_F \|B\|_F.$$

[Hint: Let $\mathbf{b}_j \in \mathbb{C}^m$ denote the j th column of B , so that $A\mathbf{b}_j$ is the j th column of AB . Now use part (b) and the column formula from part (a).]

Challenge Problem (Geometric Series of Matrices). Let A be a square matrix with $\|A\|_F < 1$. In this problem you will show that $I - A$ is invertible, with a power series expansion that converges with respect to the Frobenius norm:¹

$$(I - A)^{-1} = I + A + A^2 + \cdots = \sum_{k \geq 0} A^k.$$

- (a) Show that $\|A^n\|_F \leq \|A\|_F^n$. Use this to show that A^n converges to the zero matrix.
 (b) Let $S_n = \sum_{k=0}^n A^k$. If $m < n$, show that $\|S_n - S_m\|_F \leq \sum_{k=m+1}^n \|A\|_F^k$. From the convergence of the scalar geometric series, this proves that S_n is a Cauchy sequence, hence it converges to some matrix T .
 (c) Observe that $S_n(I - A) = (I - A)S_n = I - A^{n+1}$. Use (a) to show that the right side converges to I and use (b) to show that $S_n(I - A)$ and $(I - A)S_n$ converge to $T(I - A)$ and $(I - A)T$, respectively. Hence $T(I - A) = (I - A)T = I$, as desired.
 (d) **Application: Absorbing Markov Chains.** Consider a partitioned matrix

$$M = \left(\begin{array}{c|c} I & 0 \\ \hline R & Q \end{array} \right),$$

where I is an identity matrix, R is any rectangular matrix and Q is a square matrix satisfying $\|Q\|_F < 1$. Use the geometric series to show that

$$M^n \rightarrow \left(\begin{array}{c|c} I & 0 \\ \hline (I - Q)^{-1}R & 0 \end{array} \right) \quad \text{as } n \rightarrow \infty.$$

[Hint: Compute the first few powers of M and observe a pattern.]

- (e) **Example: Gambler's Ruin.** Consider a game where at any stage you have d dollars for some $d \in \{1, 2, 3, 4\}$. The game consists of repeatedly flipping a coin that lands heads with probability p and tails with probability $q = 1 - p$. Each time you get heads you gain a dollar and each time you get tails you lose a dollar. The game ends when $d = 1$ or $d = 4$. Consider the matrix $M = (m_{ij})$, where m_{ij} is the probability that $d = j$ next turn, assuming that $d = i$ this turn, and the matrix $P = (p_{ij})$, where p_{ij} is the probability that the game ultimately ends with $d = j$, assuming that it started with $d = i$. It follows from basic rules of probability that

$$P = \lim_{n \rightarrow \infty} M^n.$$

Use part (d) to compute the entries of P . [Hint: Reorder the states as 1, 4, 2, 3.]

¹To be precise, for any sequence of matrices $X_1, X_2, \dots$ and for any matrix Y , we say that X_n converges to Y if and only if $\|X_n - Y\|_F$ converges to the number zero. It follows from the completeness of the complex numbers that if $\|X_n - X_m\|_F$ gets arbitrarily small for n and m arbitrarily large (i.e., if X_n is a *Cauchy sequence*) then there exists some matrix Y such that $X_n \rightarrow Y$.